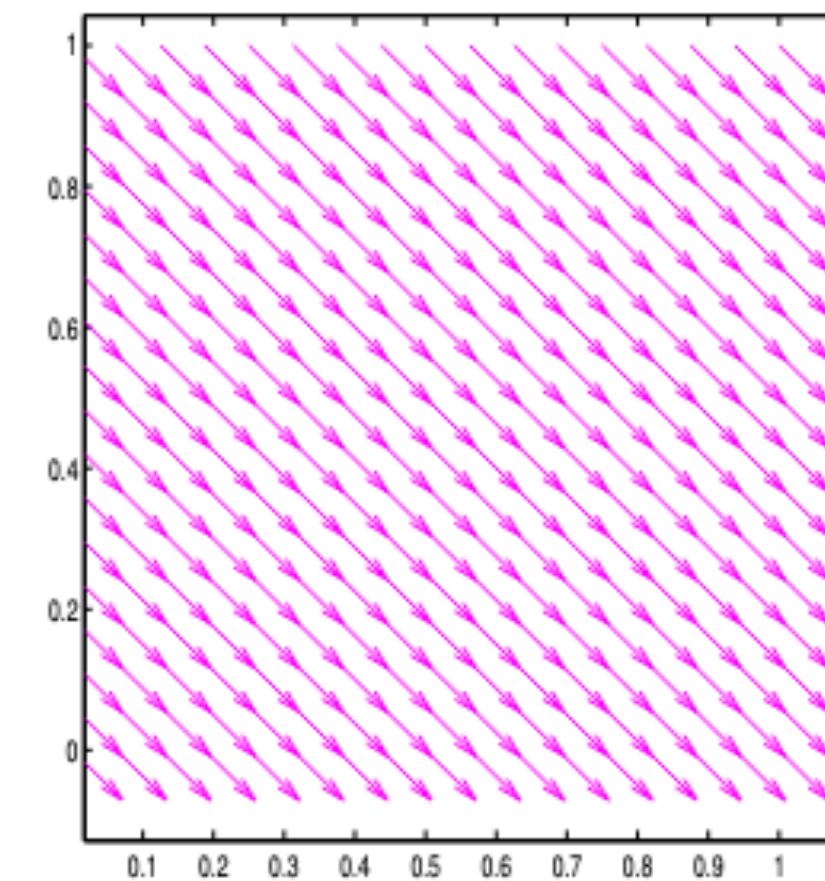
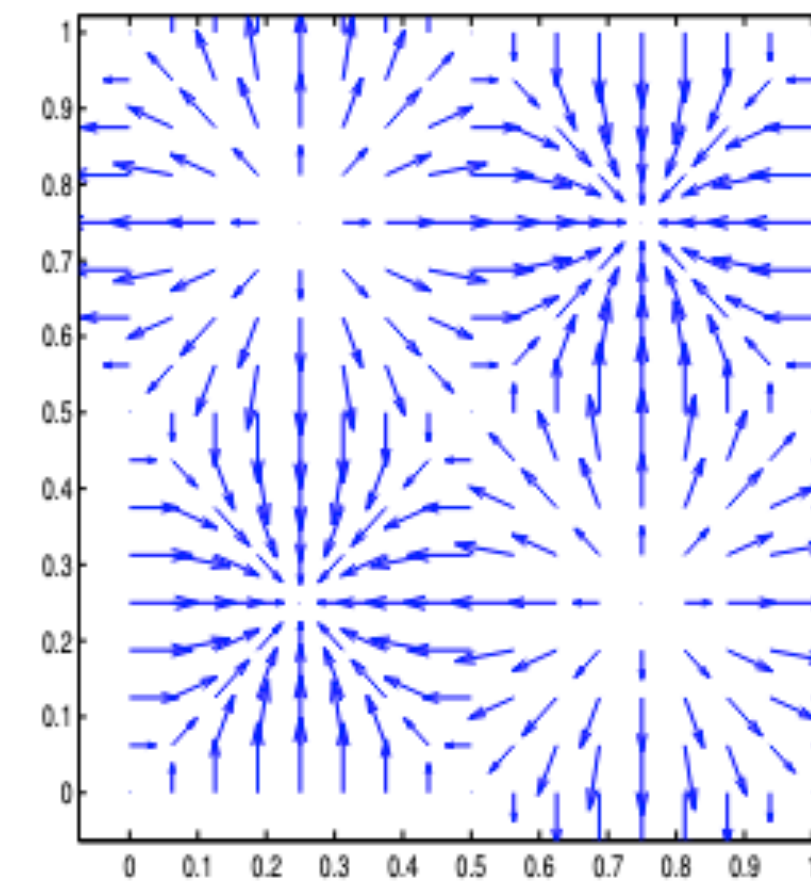
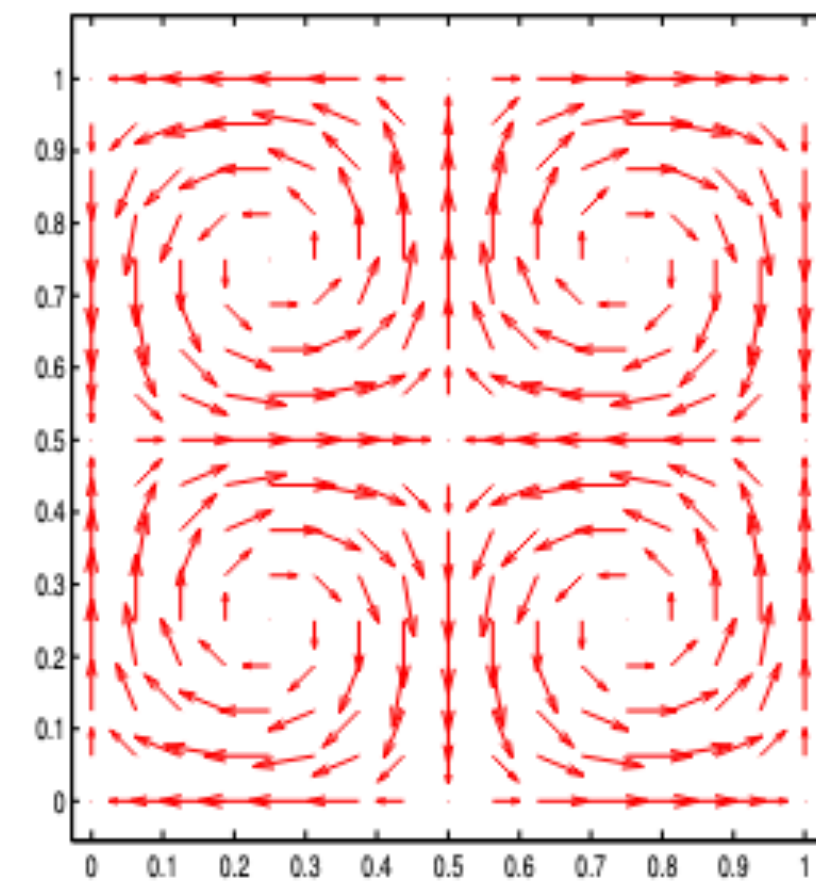
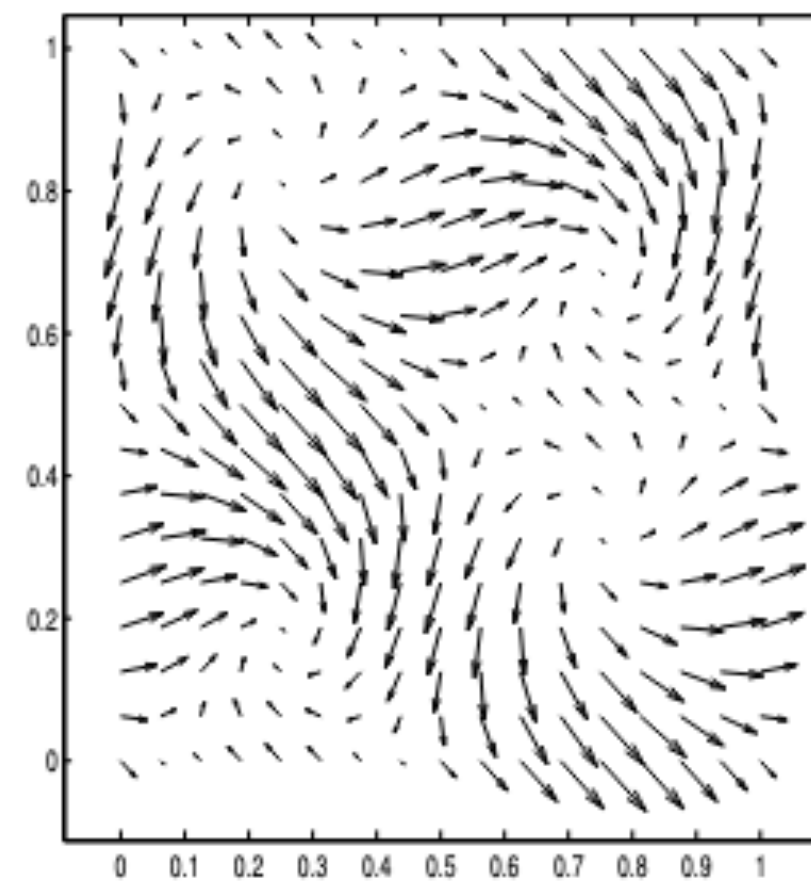


(Helmholtz) Hodge decomposition

$$\text{Edge flow} = \begin{array}{c} \text{Curl flow} \\ \text{Div-free, solenoidal} \end{array} + \begin{array}{c} \text{Gradient flow} \\ \text{Curl-free, irrotational} \end{array} + \begin{array}{c} \text{Harmonic flow} \\ \text{Div- and curl-free} \end{array}$$

Edge flows behave like vector fields

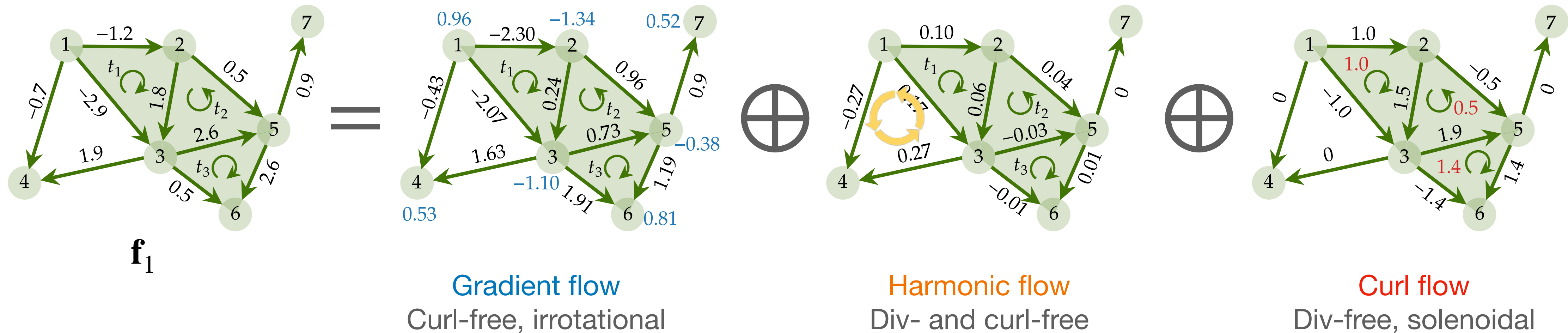


Hodge decomposition

Lovász et al. 2004; Lim et al. 2020

$$\mathbb{R}^{N_1} = \text{im}(\mathbf{B}_1^T) \oplus \text{ker}(\mathbf{L}_1) \oplus \text{im}(\mathbf{B}_2)$$

$$\mathbf{f}_1 = \mathbf{f}_G + \mathbf{f}_H + \mathbf{f}_C$$



- This holds for the space of nodes, edges, triangles ... and **almost everywhere**
- What is the case for nodes? The harmonic space $\sim$ #connected components

$$\mathbb{R}^{N_0} = \text{im}(\mathbf{B}_1) \oplus \text{ker}(\mathbf{L}_0)$$